



Clinical Application of Augmented Reality in Urology

Jian-ri Li, MD, PhD

Department of Urology, Taichung Veterans General Hospital

Disclaimer:

Invited speakers:

Urothelial cancer:

Ono, MSD, Roche, AstraZeneca, Janssen, Astellas, Merck

Prostate cancer:

Janssen, Astellas, Amgen, Ipsen, AstraZeneca, Bayer, Ferring, Synmosa

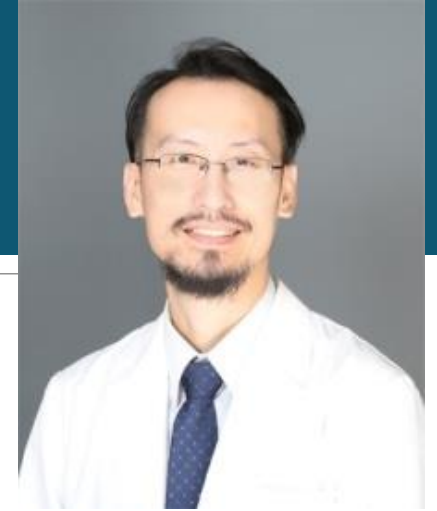
Kidney cancer:

Pfizer, Novartis, Ipsen, Ono, MSD

Voiding dysfunction:

Synmosa, Astellas, Pfizer, GSK, Boston Scientific, Teleflex

About Dr. Jian-ri Li



Chair, Department of Urology, Taichung Veterans General Hospital

Professor, School of Medicine, National Chung Hsing University

Clinical Practice

Robotic surgeries in uro-oncology

Endourology in prostate surgery

Research Interests

Biomarkers in urological cancers

How we expect for the future surgery

KIA	RIANNON, STEVIE	KIA	ROSE
WIA	RIBICON, WELLER	MIA	ROSS
KIA	RIBST, CARLOTTA	KIA	ROSS
MIA	RICARDO, NYE	KIA	ROTE
K	RICO, JOHN D.	KIA	JTH
M			JTH
K			JTH
K			JTH
K			JTH
K			JTH
KIA	RICKER, ETHAN	MIA	ROTI.
KIA	RICO, JOHN D.	KIA	ROTO
MIA	RICOS M	WIA	ROTO

COMMENTARY

Emerging technology can replace workers — or train them for new work

Jack Karsten
August 29, 2019



Application of AR in Urological Surgeries

Reconstructive surgeries

- Complicated renal stones

- Ureteral reconstruction

Oncology (Tumor margin)

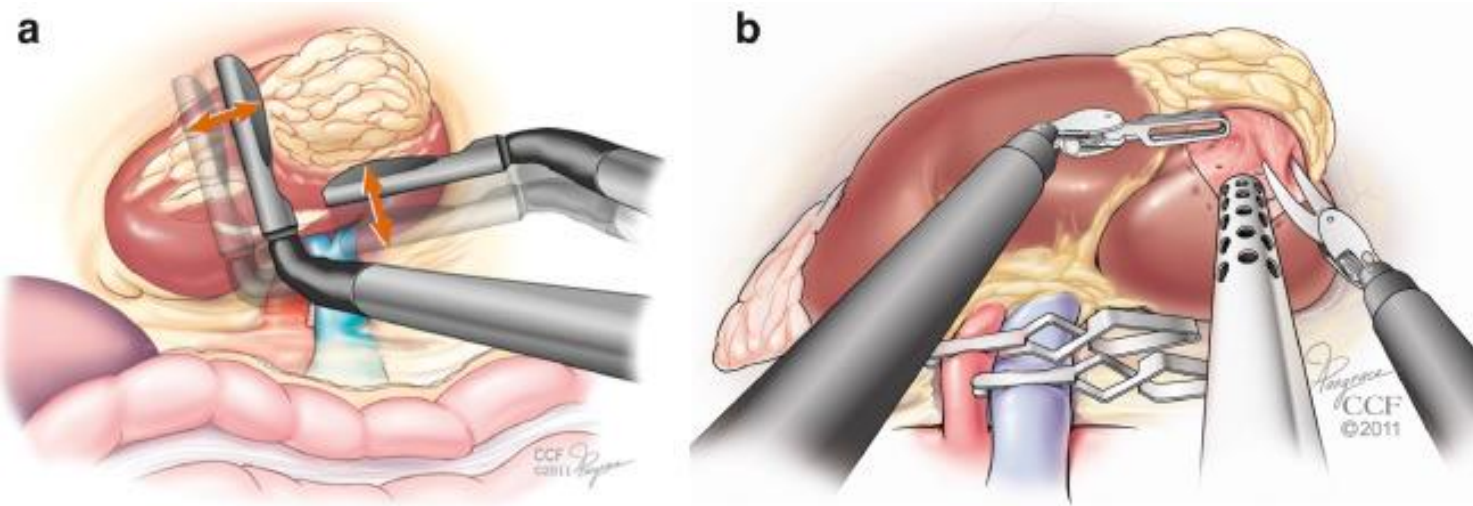
- Partial nephrectomy

- Partial cystectomy

- Radical prostatectomy

- Poorly identified retroperitoneal tumor

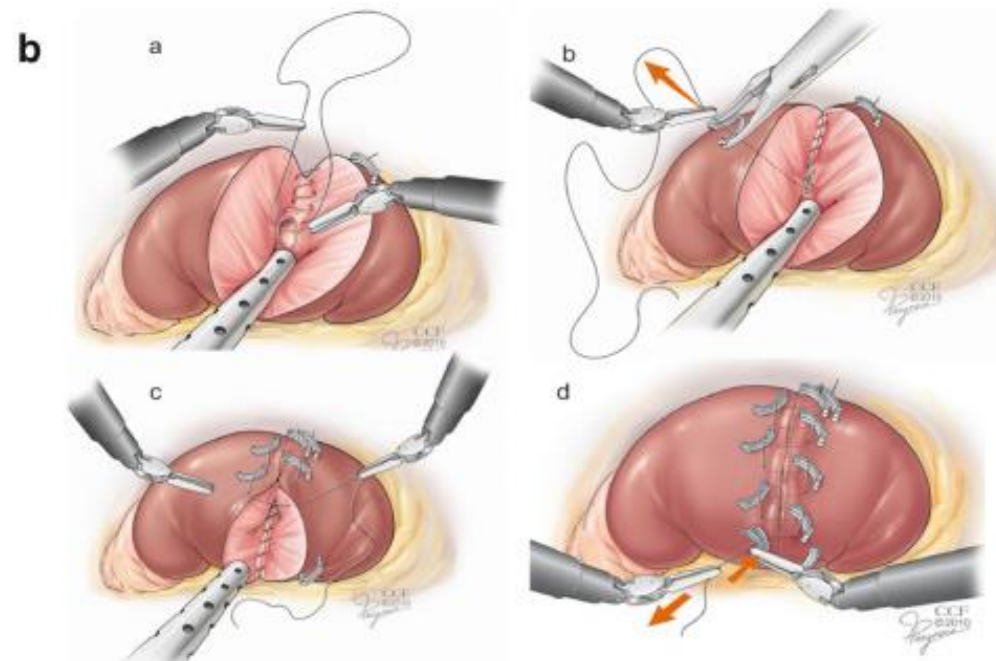
Partial nephrectomy is challenging for Urologists



Tumor identification and resection

Reconstruction

On time



Methods to improve tumor identification

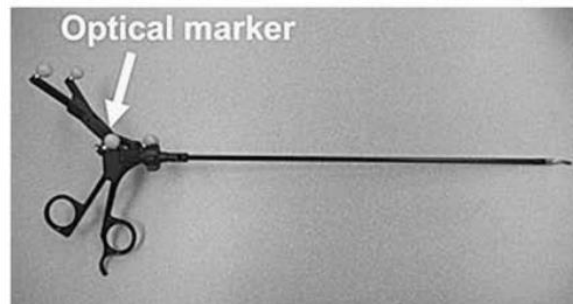
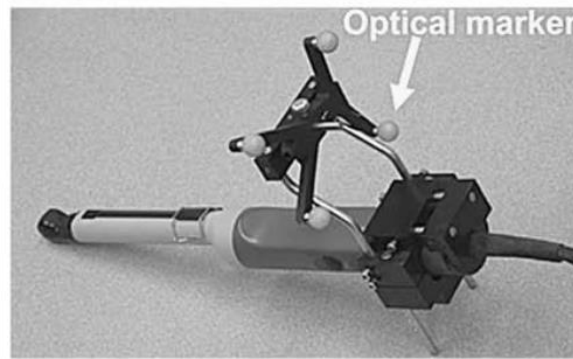
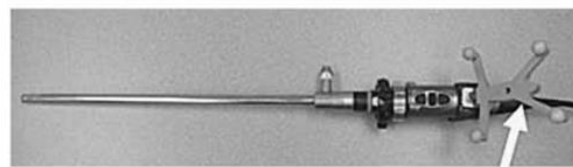
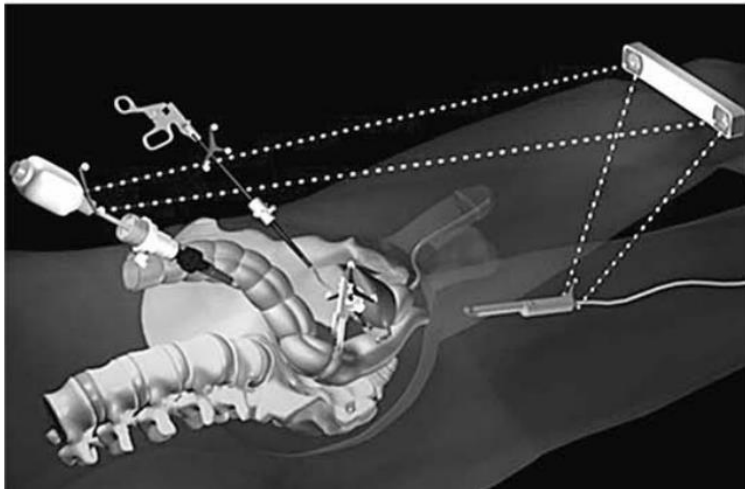
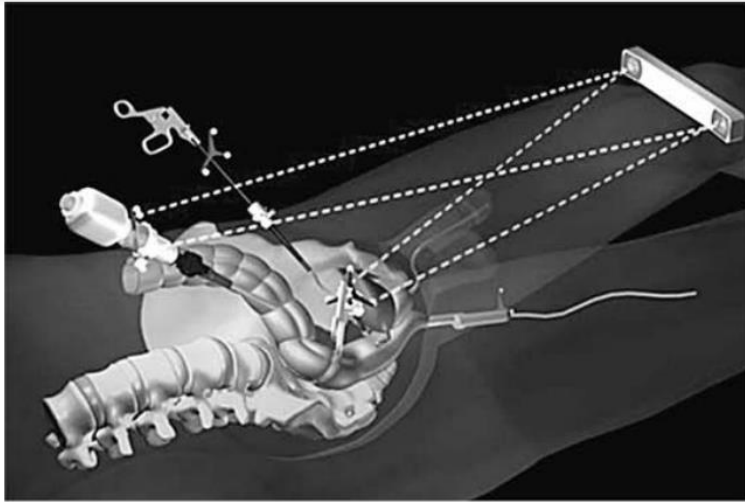
Intra-operative ultrasound

Pre-operative 3D reconstruction

Intra-operative Indocyanine green

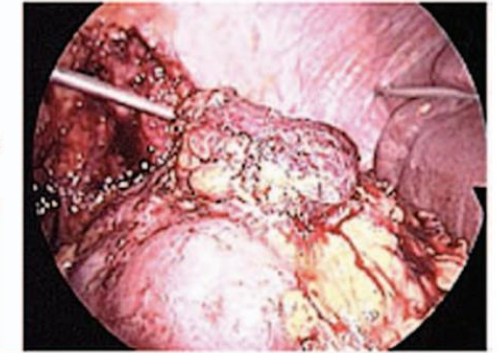
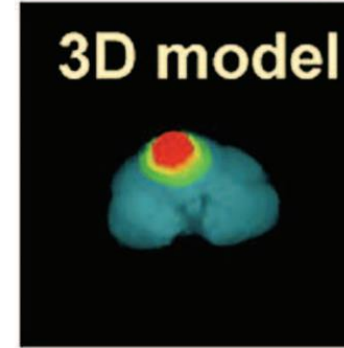
Augmented Reality

First AR practice



Color-coded Zonal Navigation

- Red zone : Tumor
- Yellow zone: 0-5 mm margin
- Green zone: 5-10 mm margin
- Blue zone: Over 10 mm margin



Methods to improve tumor identification

	Ultrasound	Pre-op 3D	ICG	AR
Tumor location	++	+++	+++	+++
Vascular structure	+	+++	+	+++
Collecting system correlation	+	+	+	++
Dynamic	+	+	++	++

Save

Email

Send to

Sort by:

Publication date



Display options



MY NCBI FILTERS

52 results



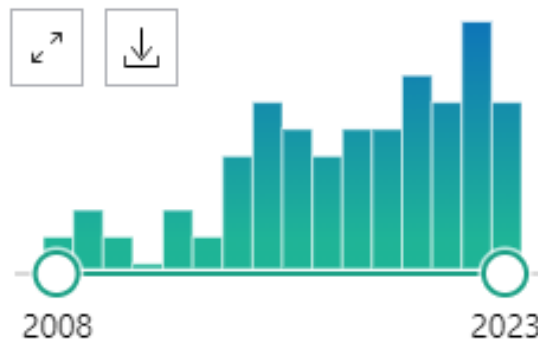
Page

6

of 6



RESULTS BY YEAR



TEXT AVAILABILITY



Abstract



Free full text



Full text



51

Cite

Share

Augmented reality during robot-assisted laparoscopic **partial nephrectomy**: toward real-time 3D-CT to stereoscopic video registration.

Su LM, Vagvolgyi BP, Agarwal R, Reiley CE, Taylor RH, Hager GD.

Urology. 2009 Apr;73(4):896-900. doi: 10.1016/j.urology.2008.11.040. Epub 2009 Feb 4.

PMID: 19193404

OBJECTIVES: To investigate a markerless tracking system for real-time stereo-endoscopic visualization of preoperative computed tomographic imaging as an **augmented** display during robot-assisted laparoscopic **partial nephrectomy**. METHODS: Stereoscopic video segm ...



52

Cite

Share

Imaging-assisted endoscopic surgery: Cleveland Clinic experience.

Ukimura O, Gill IS.

J Endourol. 2008 Apr;22(4):803-10. doi: 10.1089/end.2007.9823.

PMID: 18366316 Review.

Augmented reality was feasible and useful to facilitate the surgeon's direct interpretation of 3D

Obstacles in Evolving AR

Time consuming in 3D
slicing pre-op

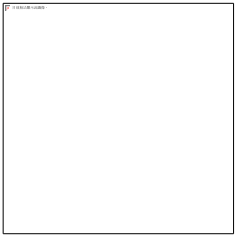
Dynamic change during
operation

No Benefit

No extra charge

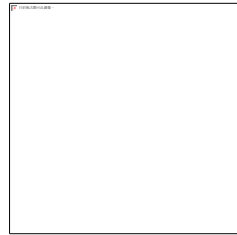
Additional workload

Things Changes



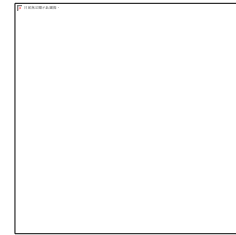
3D technology

3D image programming improving



Fusion technique

Real-time technique improving



Hospital support

Intelligent development



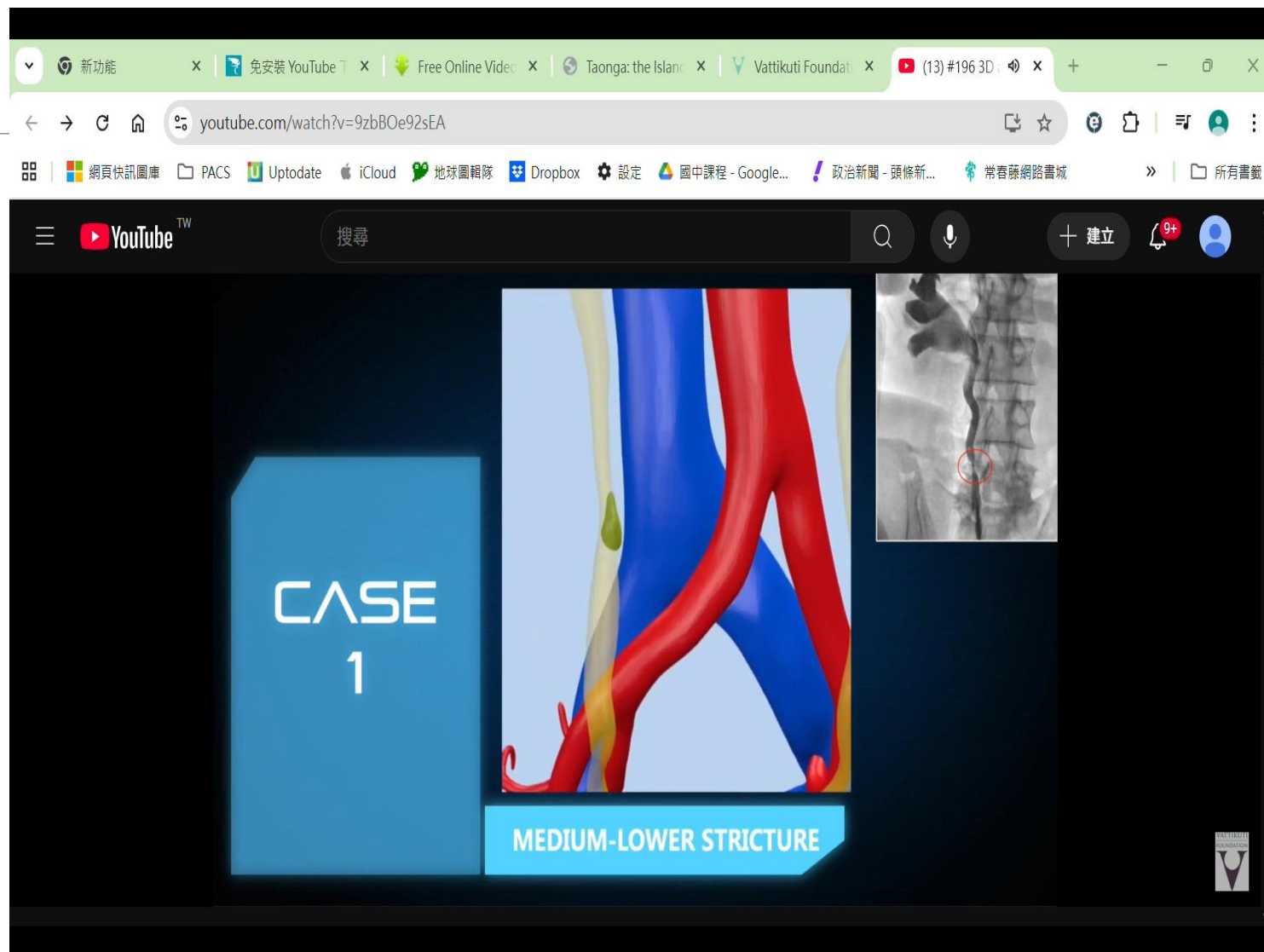
Augmented reality 3D robot-assisted partial nephrectomy: tips and tricks to improve surgical strategies and outcomes

*F. Piramide, D. Amparore, A. Pecoraro, P. Verri, E. Checcucci, S. De Cillis,
A. Piana, M. Manfredi, C. Fiori, F. Porpiglia*



Division of Urology
SCHOOL OF MEDICINE – TURIN
“S. Luigi Gonzaga Hospital” - Orbassano (Turin), Italy





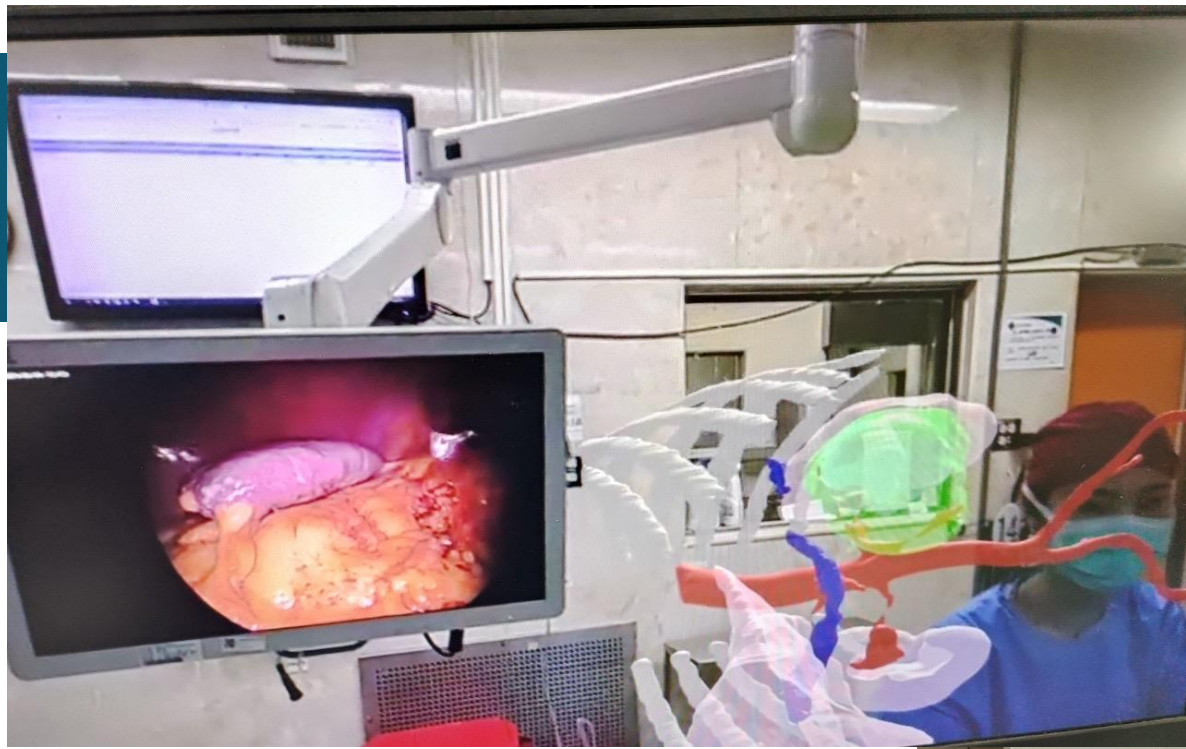
Surgeon Training Program in Taichung Veterans General Hospital

First-year attending urologist

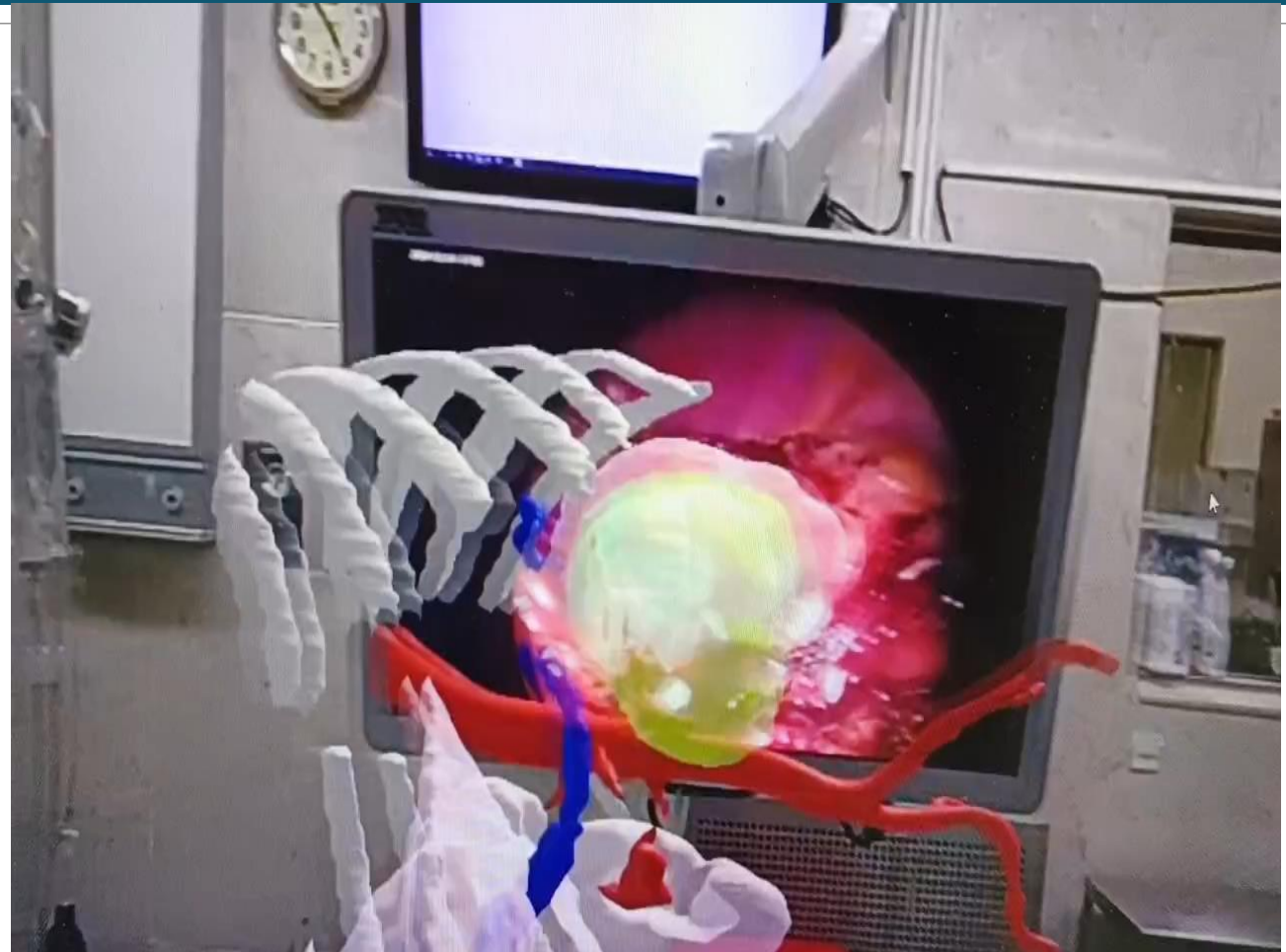
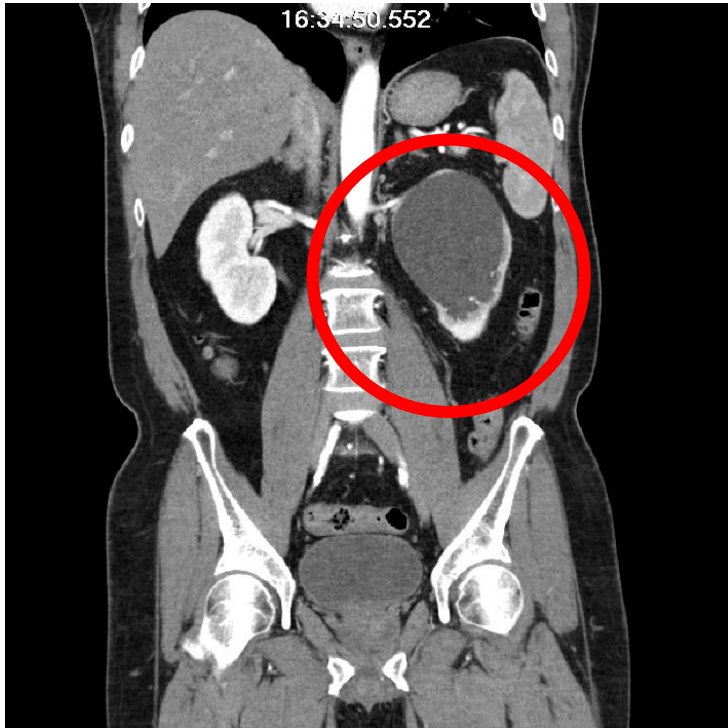
Experience of assistant on laparoscopic or robotic surgeries for more than 100 cases

Experience of laparoscopic surgery for more than 10 cases

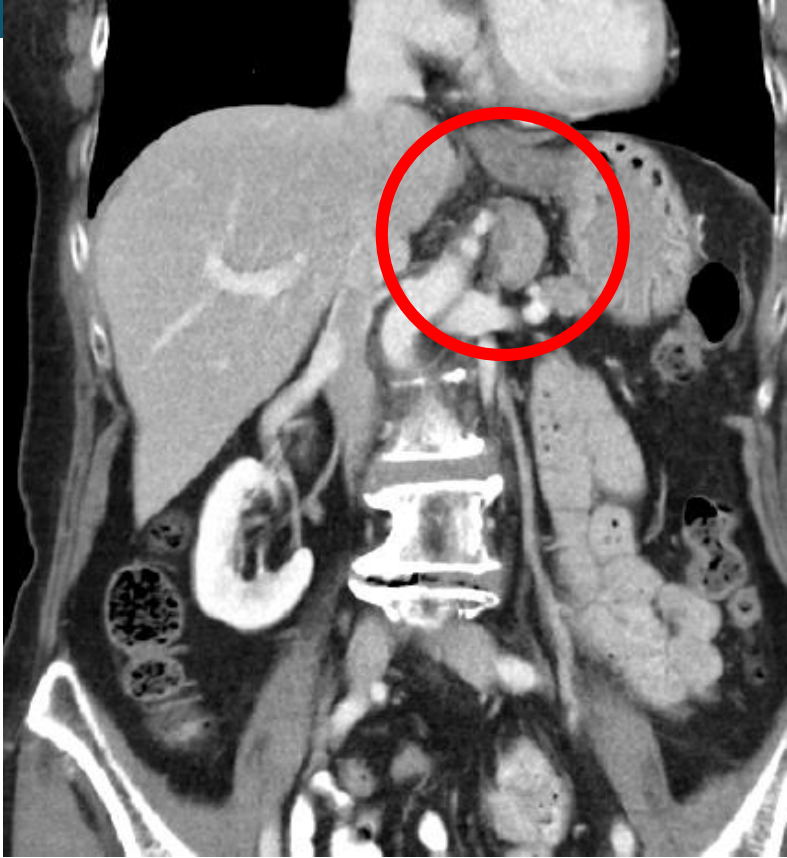
Comparison of AR assistant surgical approach and conventional



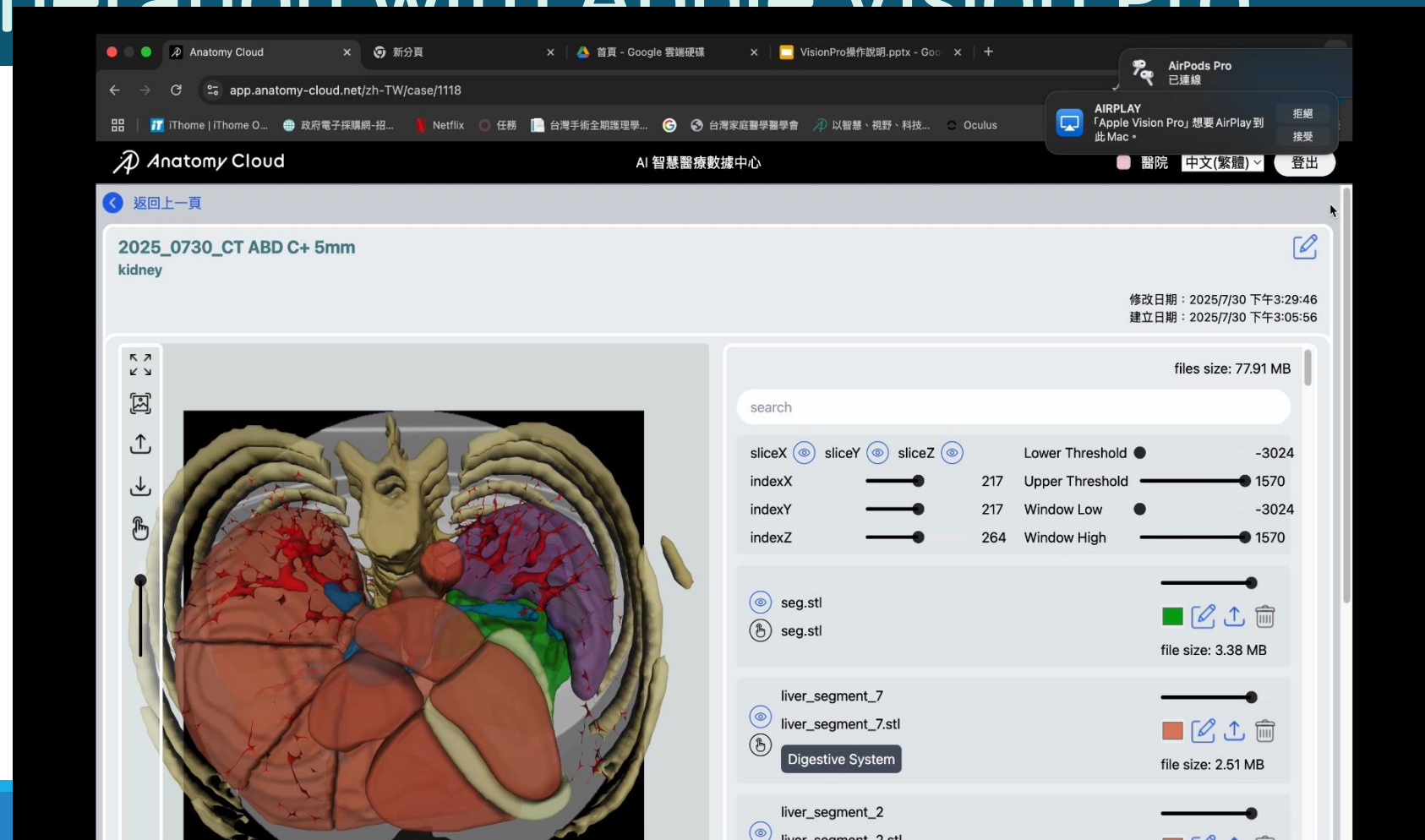
Left renal pelvis reconstruction



Left supra-renal paraganglioma



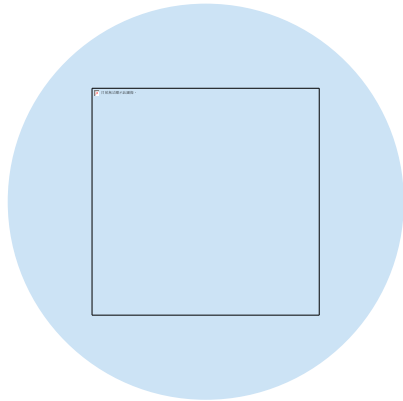
Second generation with Apple Vision Pro



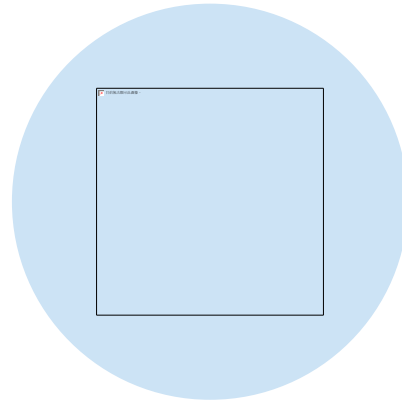
Comparison between AR and Conventional laparoscopic surgery

	AR assisted	Conventional laparoscopy
Patient parameters		
Number	5	5
Age	61	58
BMI	26.3	25.6
Side position (R/L)	3/2	2/3
Surgery		
Kidney	3	2
Supra-renal	2	3
Surgeon parameters		
Approach time (minutes)	32.3	41.2
Satisfaction	4	3

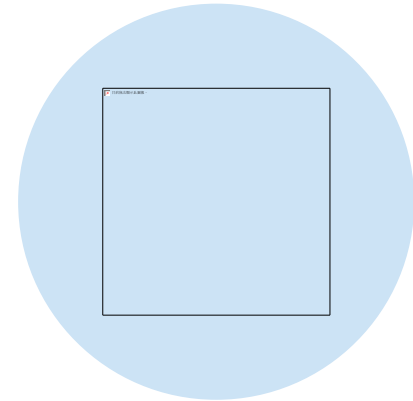
Benefit of Current Augmented Reality Assisted Surgeries



ASSISTANT REAL-TIME
IDENTIFICATION



TRAINING AND
EDUCATION



QUALITY CONTROL

Challenges of Current AR assisted Surgery

Real-Time is
not real

Pre-operative
image
preparation

Education only



Thanks for your attention
